

PC Maintenance, Security and Troubleshooting:

Overview of Computer Maintenance and Security

Computer Maintenance involves routine tasks to keep hardware clean, software optimized, and performance smooth.

Security protects against threats like viruses, hackers, data theft, and ransomware.

Key goals:

- Prevent slowdowns, crashes, and data loss
- Protect personal information (banking, passwords, photos)
- Extend hardware life

Inbuilt PC Security Tools (Windows 11)

Windows has excellent built-in security — no need for third-party tools in most cases.

Main tool: **Windows Security** (formerly Windows Defender)

- Virus & threat protection
- Firewall & network protection
- App & browser control
- Device security
- Family options

Here are examples of the **Windows Security** dashboard (showing virus protection, firewall status, and quick scan options):

(Imagine clean, modern interface with green checkmarks for protected status.)

It runs real-time protection automatically. Run **Quick Scan** or **Full Scan** weekly.

Securing Documents

- Use strong passwords for files (right-click file → Properties → Advanced → Encrypt contents).
- Store important files in **OneDrive** with Personal Vault (requires 2FA).
- Avoid saving sensitive data (Aadhaar, bank details) on Desktop or Downloads folder.
- Use **BitLocker** (built-in) to encrypt entire drive on laptops (Settings → Privacy & security → Device encryption).

Antivirus

Windows Security (Microsoft Defender) is sufficient and free for most home users in 2026 — it's lightweight, always updated, and highly effective against modern threats.

If you want extra protection (e.g., for online banking), consider reputable paid options like Bitdefender, Kaspersky, or Norton — but avoid unknown free antiviruses (many are fake/malware).

Run scans regularly and keep definitions updated.

Here are examples of an **antivirus scan in progress** (progress bar, detected threats list, and clean result screen).

Upgrading Operating System and Application Software

Regular updates fix security holes and improve performance.

- **Windows Update** → Settings (Win + I) → Windows Update → Check for updates.

Enable automatic updates — install security patches immediately.

Here are examples of the **Windows Update** settings screen (showing update history, active hours, and restart options).

- Update apps via **Microsoft Store** (Library → Get updates).
- For other software (browsers, Adobe, etc.), use built-in update checkers or tools like Patch My PC.

Cleaning the Monitor, Keyboard, CPU

Important: Always power off and unplug the computer first!

- **Monitor** → Use microfiber cloth + distilled water or screen cleaner spray (never household cleaners). Wipe gently in circular motion.
- **Keyboard** → Turn upside down and shake gently. Use compressed air to blow dust. For sticky keys, use isopropyl alcohol (70%) on cotton swab (not dripping wet).
- **CPU / Cabinet** → Open side panel → Use compressed air to blow dust from fans, heatsink, and vents (important for cooling). Clean every 6-12 months.

Here are safe cleaning step-by-step visuals (microfiber cloth on screen, compressed air on keyboard/fans, and cleaned internals).

Deleting Unnecessary Programs and Files

- **Uninstall Programs** → Settings → Apps → Installed apps → Uninstall unused software.
- **Disk Cleanup** → Search for "Disk Cleanup" → Select drive (usually C:) → Check Temporary files, Recycle Bin, etc. → OK → Clean up system files for more space.

Here are examples of **Disk Cleanup** tool in Windows (file selection screen and cleanup progress).

- **Delete Toolbars / Browser Extensions** → In browser (Chrome/Edge) → Extensions → Remove suspicious ones.

Defragment Hard Drive

For **HDD** (traditional hard disk): Defragmentation organizes files for faster access.
For **SSD** (most modern PCs): Do NOT defrag — Windows optimizes automatically (TRIM).

Steps:

Search for "Defragment and Optimize Drives" → Select drive → Optimize.

Here are visuals of the **Optimize Drives** tool (showing drive types, last run time, and optimization in progress).

Computer Maintenance Programs

Some popular third-party tools (use with caution — stick to official sites):

- **CCleaner** → Cleans junk files, browser cache, registry (free version is good; avoid paid upsells).
- **MyDefrag** → Advanced defragmenter (free, open-source).
- **SpinRite** → Powerful hard drive repair/recovery tool (paid, by Gibson Research).

Here are examples of the **CCleaner** main interface (cleaner tab with junk file scan results).

Tip: Windows built-in tools are safer and sufficient for most users.

Basic Troubleshooting

Quick fixes for common issues:

1. **Restart Computer** → Simplest and most effective (fixes 80% of problems).
2. **Check Cables** → Ensure power, monitor, keyboard, mouse cables are secure.
3. **Uninstall Problem Software** → If issue started after installing something → Apps → Uninstall.
4. **Start Windows in Safe Mode** → Restart while holding Shift → Troubleshoot → Advanced options → Startup Settings → Restart → Choose 4 (Safe Mode).
Great for removing malware or fixing driver issues.

Here are examples of **Safe Mode** boot screen (black background with Safe Mode text in corners).

Follow these steps monthly for best results. Your computer will run faster, safer, and last longer! If you face any specific issue (slow PC, virus suspicion), describe it and I can guide further, Sam. 😊